

2(a)(i)	$pV = NkT$ or $pV = nRT$ <u>and</u> $N = nN_A$ $N = \frac{2.3 \times 10^5 \times 3.5 \times 10^{-3}}{1.38 \times 10^{-23} \times 294}$	C1
	$= 2.0 \times 10^{23}$	A1
2(a)(ii)	$pV = \frac{1}{3} Nmc^2$ $c^2 = \frac{3 \times 2.3 \times 10^5 \times 3.5 \times 10^{-3}}{2.0 \times 10^{23} \times 40 \times 1.66 \times 10^{-27}}$ $= 182\,000$ r.m.s. speed = 430 m s ⁻¹	C1
	or $\frac{1}{2} mc^2 = \frac{3}{2} kT$	A1
	$c^2 = \frac{3 \times 1.38 \times 10^{-23} \times 294}{40 \times 1.66 \times 10^{-27}}$ $= 183\,000$	(C1)
	r.m.s. speed = 430 m s ⁻¹	(A1)

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Question	Answer	Marks
2(b)	$c^2 = \frac{3 \times 2.0 \times 10^{23} \times 1.38 \times 10^{-23} \times (294 + 84)}{2.0 \times 10^{23} \times 40 \times 1.66 \times 10^{-27}}$ $c^2 = 236000$ $c = 485$	C1
	$ratio \left(= \frac{485}{430} \right) = 1.1$	A1
	OR $v \propto \sqrt{T}$ or $v^2 \propto T$	(C1)
	$ratio = \sqrt{\frac{273 + 21 + 84}{273 + 21}}$ or $\sqrt{\frac{378}{294}}$ ratio = 1.1	(A1)

Question	Answer	Marks
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